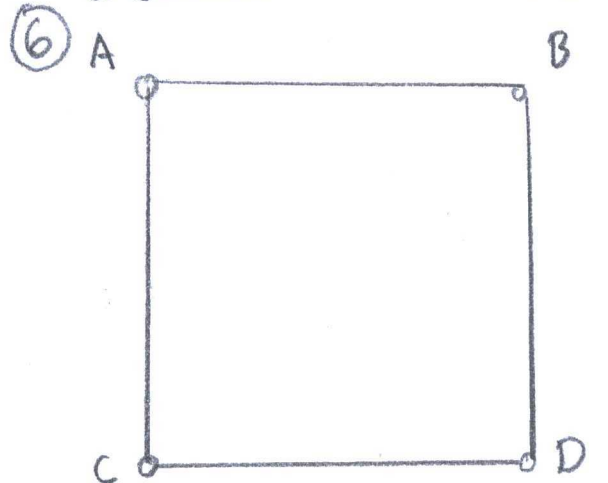
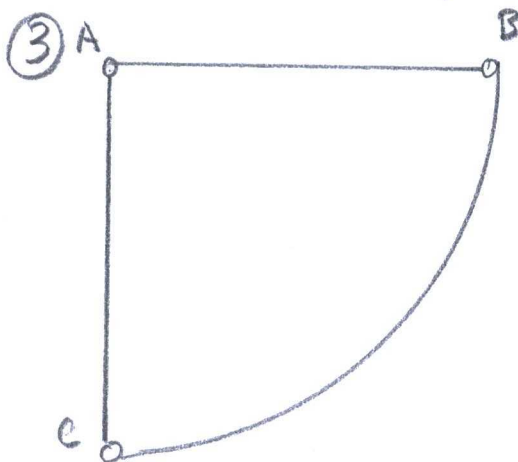
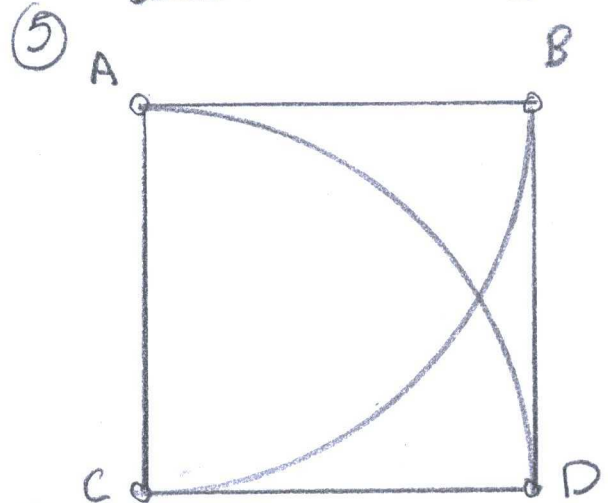
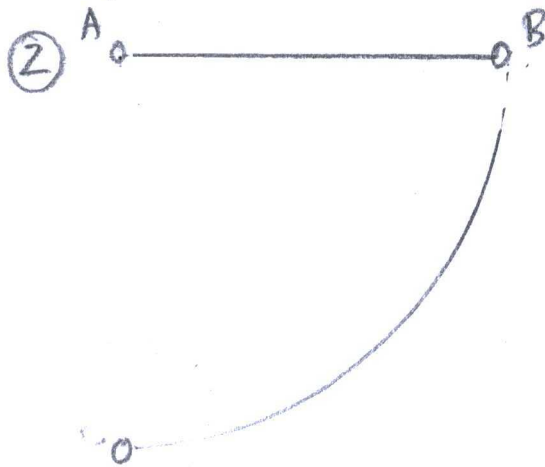
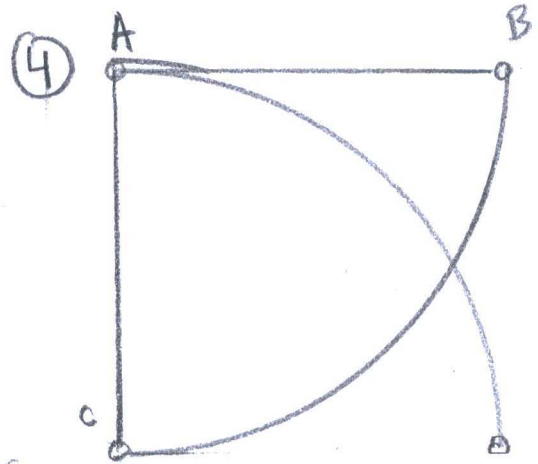


Euclidean and Non-Euclidean Geometry:

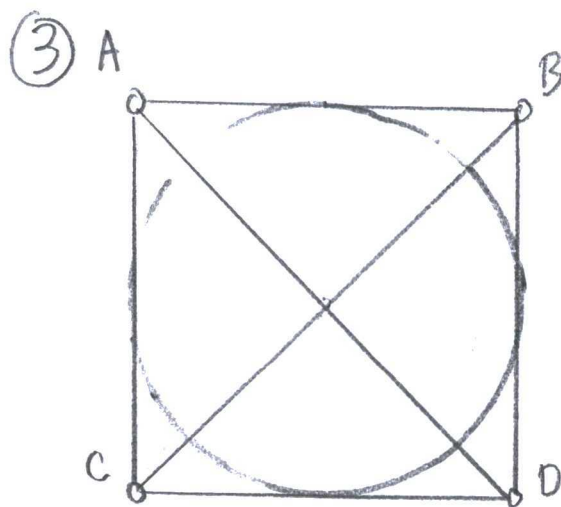
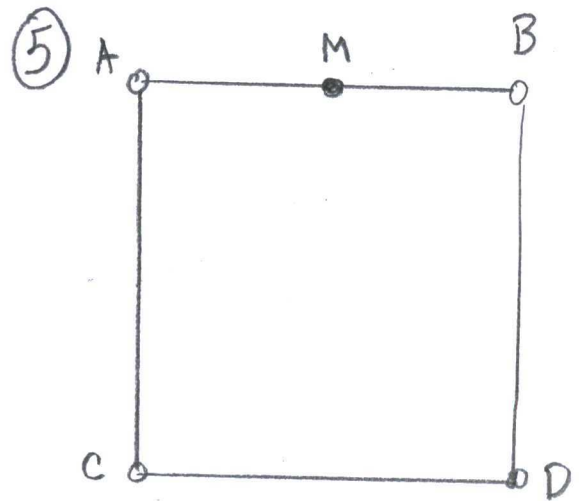
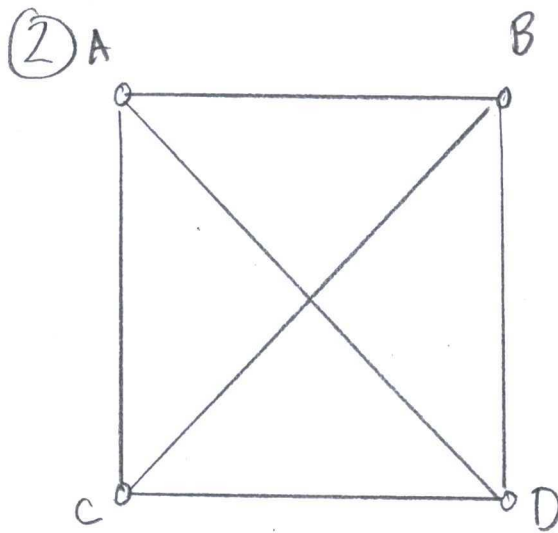
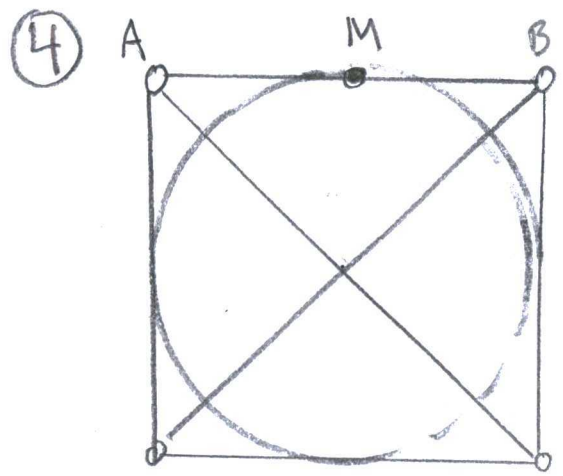
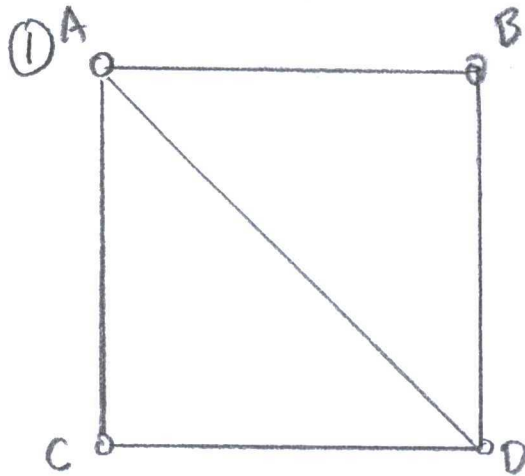
Major Exercises:

1) With straight edge and compass:

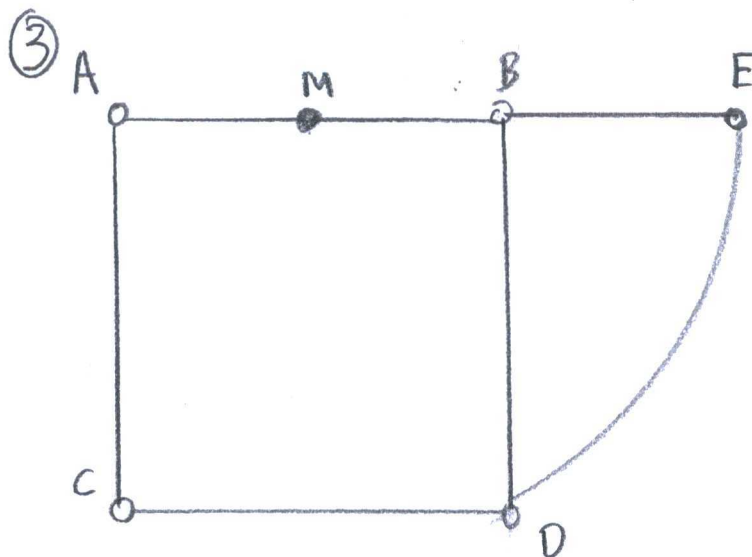
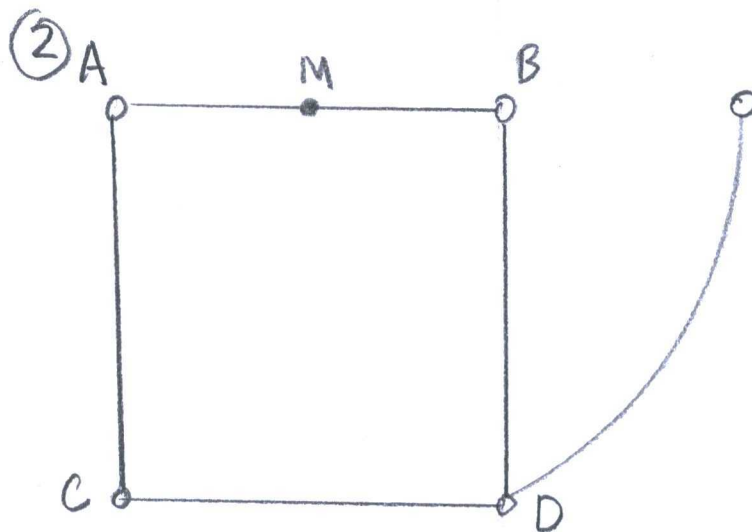
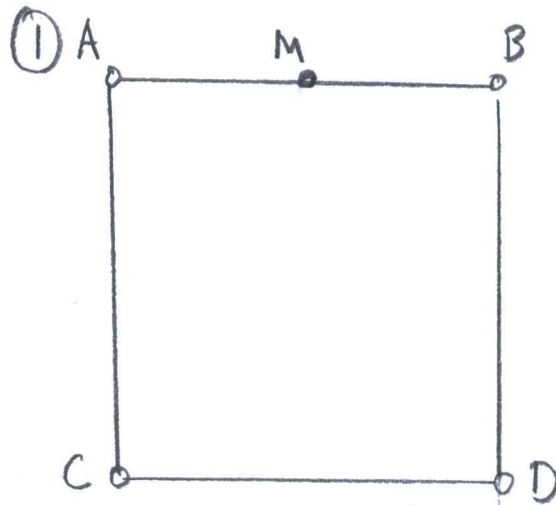
a) Construct a square $\square ABCD$

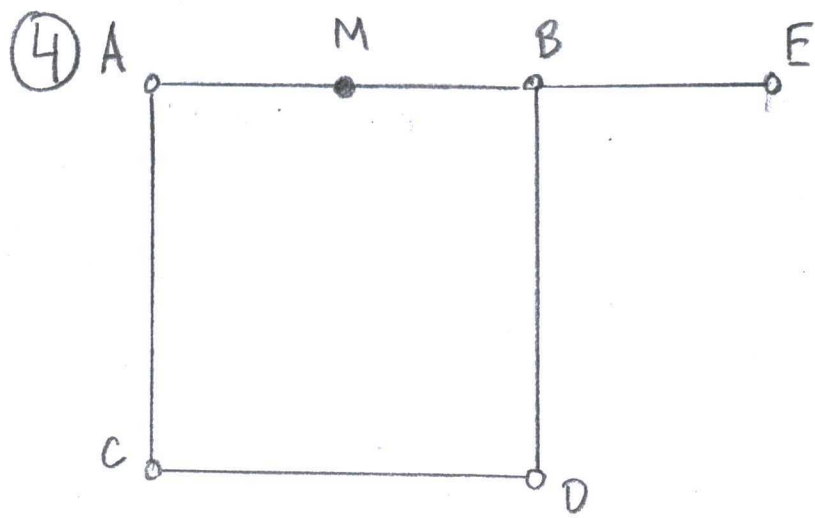


b) Construct a midpoint of AB

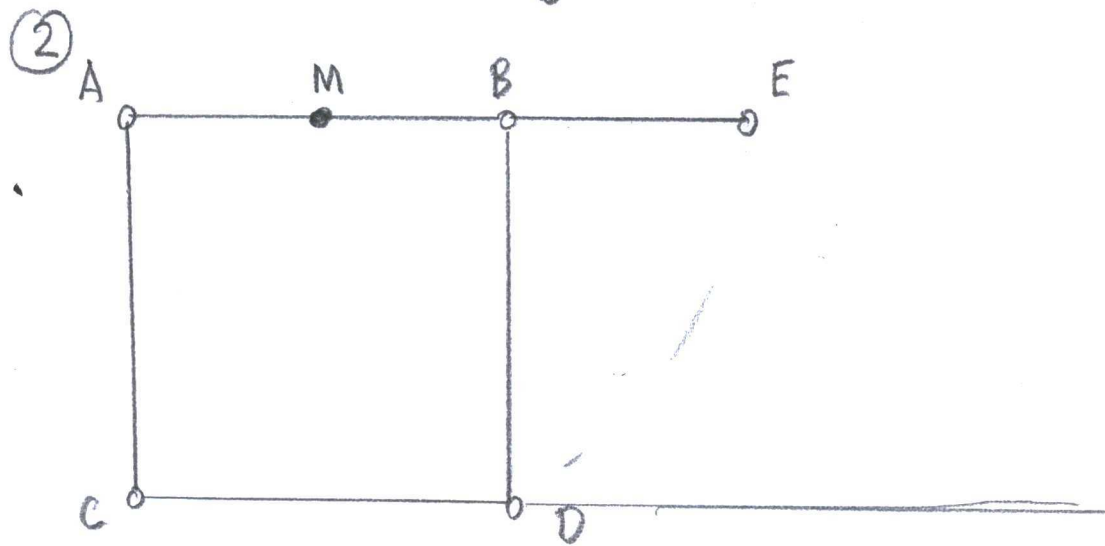
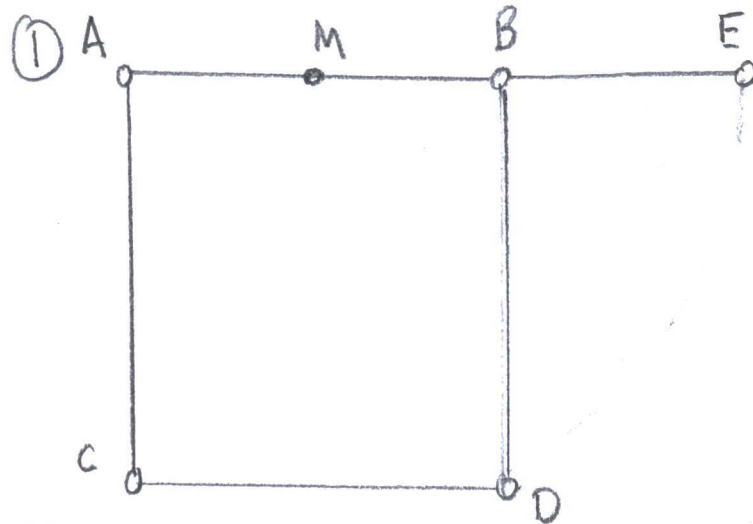


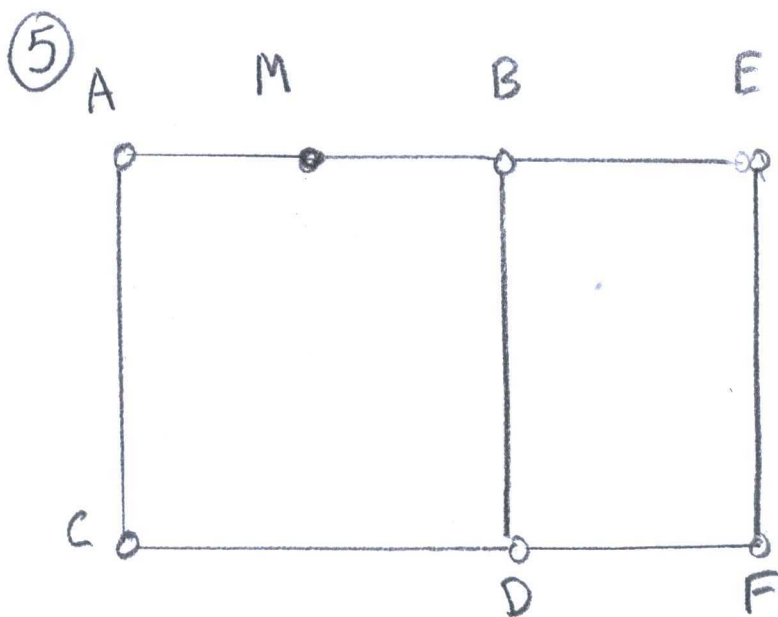
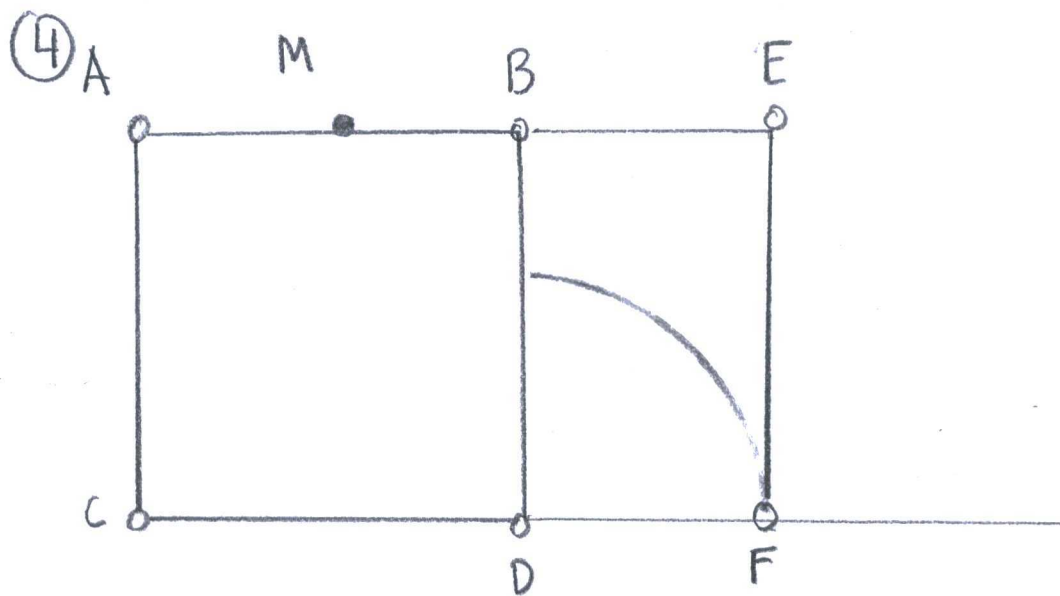
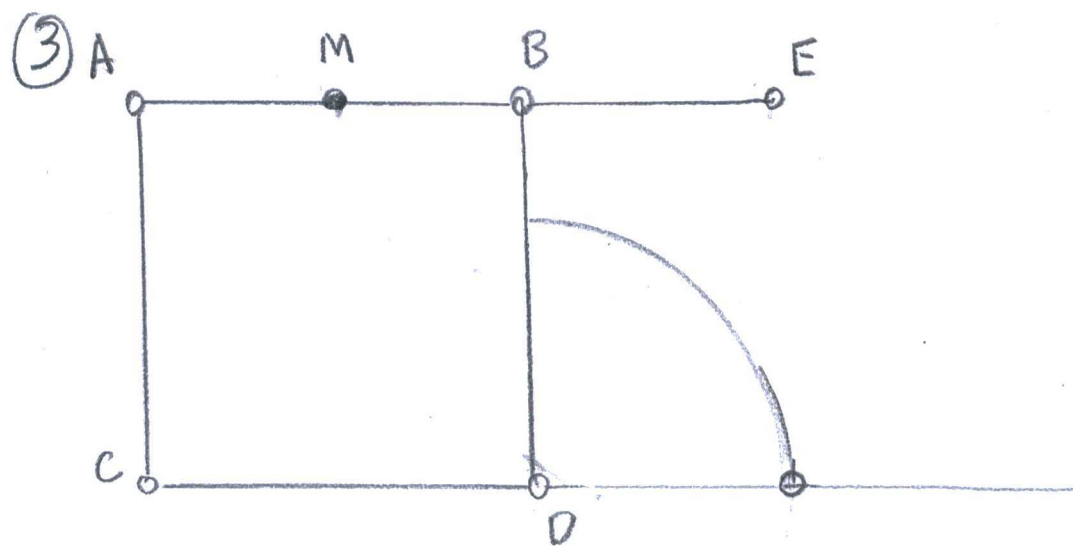
c) Construct point E such that B
is between A and E and
 $MC \cong ME$



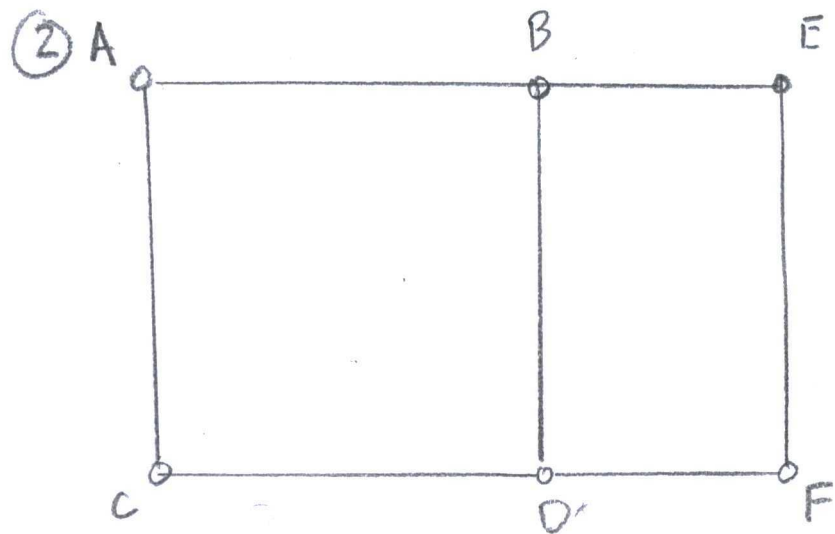
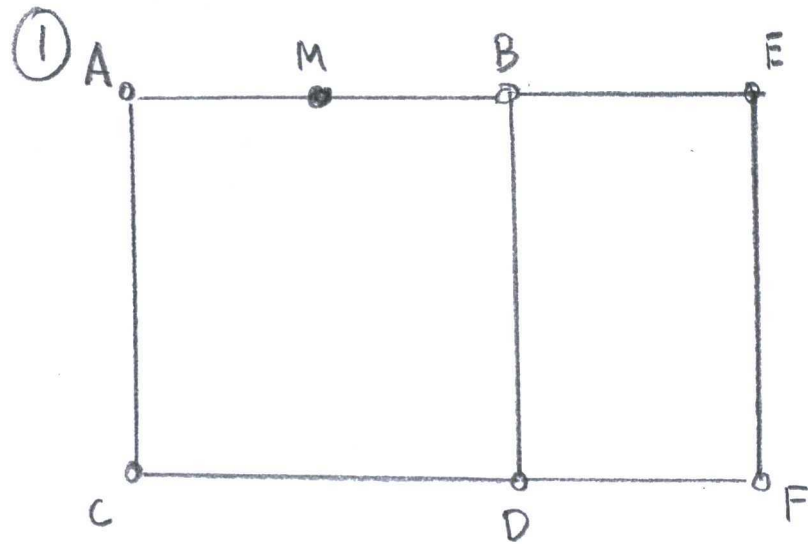


c) Construct a foot F of the perpendicular from E to DC:

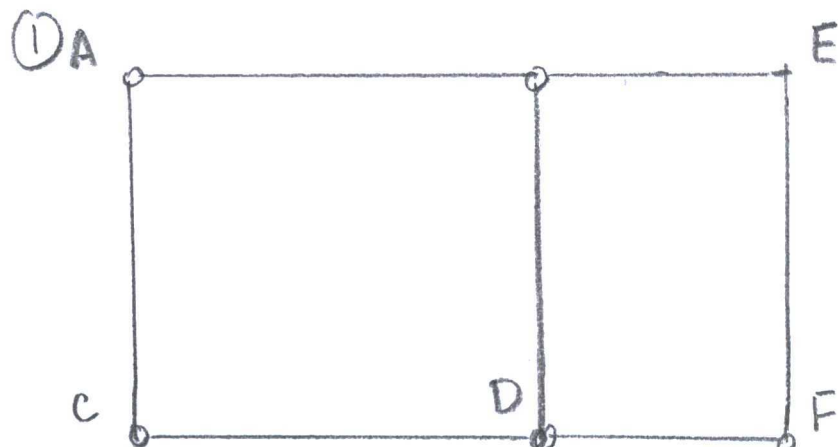




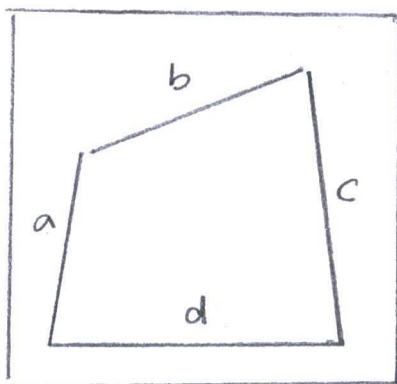
e) Then $\square AEFD$ is a golden rectangle:



f) Moreover, $\square BEDF$ is another golden rectangle:



2.



Area non-Right Triangle:

$\text{Area} = \frac{1}{2}ab\sin C$

Egyptian Area: $\text{Area} = (a+c)(b+d)/4$

My Area: $\text{Area} \leq \frac{1}{2} \left(\frac{ab\sin\theta_1}{2} + \frac{ad\sin\theta_2}{2} + \frac{bc\sin\theta_3}{2} + \frac{cd\sin\theta_4}{2} \right)$
 $= \frac{(a+c)(b+d)}{4} \text{ @ right angle}$

Authors Area: $4S \leq (a+c)(b+d)/4$

3.

(from exercise) "Formulas"

$$2S = bc \sin \theta$$

$$2bc \cos \theta = b^2 + c^2 - a^2 \text{ (law of cosines)}$$

$$\cos(60^\circ - \theta) = (\cos \theta + \sqrt{3} \sin \theta)/2$$

Identity:

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$2S = bc \sin \theta$$

$$= \frac{bc}{\sqrt{3}} (2 \cos(60^\circ - \theta) - \cos \theta)$$

$$= \frac{bc}{2\sqrt{3}} (4\cos(60^\circ - \theta) - 2\cos(\theta))$$

$$\leq \frac{1}{2\sqrt{3}} (2(b^2 + c^2) - b^2 - c^2 + a^2)$$

$$\leq \frac{1}{2\sqrt{3}} (a^2 + b^2 + c^2)$$

$$45\sqrt{3} \leq a^2 + b^2 + c^2$$

4.

I tried the problem.

Definitions/Equations necessary:

- Bisector: Half-angle
- Perpendicular Bisector: line through segment midpoint with negative reciprocal slope
- Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- General Equation: $x^2 + y^2 + Dx + Ey + F = 0$ to a circle
- Slope: $m_1 = \frac{y_2 - y_1}{x_2 - x_1}$
- Perpendicular slope: $m_2 = -\frac{1}{m_1}$
- Point-slope formula: $y - y_1 = m(x - x_1)$

Trial points:

$$A=(0,0), B=(0,2), C=(1,0)$$

Circle through points:

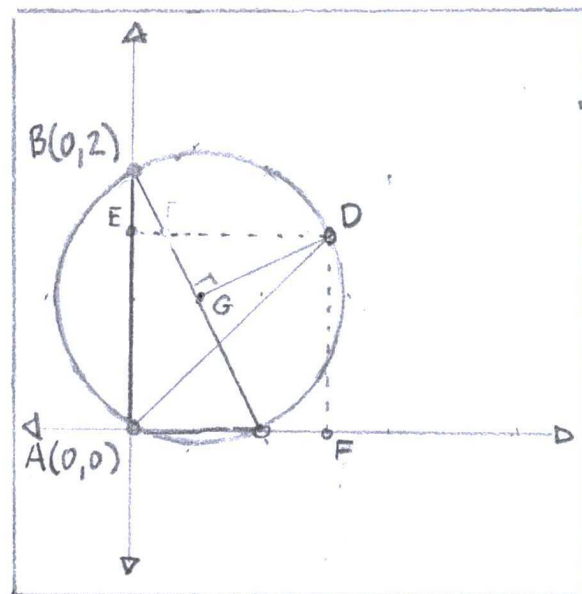
$$(x - \frac{1}{2})^2 + (y - 1)^2 = 5/4$$

Perpendicular bisector:

$$y = \frac{1}{2}x + \frac{3}{4}$$

Bisector:

$$y = x$$



(a) Yes, D lies outside the triangle on a circle through ABC.

(b) Yes, one of E or F lies inside the triangle and the other outside.

(c) Lastly, yes! E, F, and G are collinear from the line through $(0, \frac{3}{2})$, $(\frac{3}{2}, 0)$, and $(\frac{1}{2}, 1)$, further $y = -x + \frac{3}{2}$.

5.

(from exercise) "Stymie"

"If a natural number n is not the square of some other natural number, then $\sqrt{n}$ is irrational."

Citation: Mazur, B.

"How did Theaetetus prove his theorem?"

Harvard

Basic Outline:

① An array containing the square root of natural numbers without perfect squares: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \dots$

② Loop through integers (x)

and for each integer loop through further ints (y)

(a) Check if the array has $\sqrt{n} = x/y$

(b) Two methods exist if in array

i) Theaetetus said, "If $\sqrt{n}$ is irrational, so is $\sqrt[3]{n}$ "

ii) Mazur said, "If $\sqrt{n}$ is in array, double check $n = y^2 = x^2$ by computation."